DATA ENGINEERING

IMPORTANT CONCEPTS TO KNOW

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DATA WAREHOUSING

Centralized storage that combines data from multiple sources, designed for query and analysis.

Building a data warehouse using Snowflake to integrate and store sales, marketing, and CRM data.



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ETL / ELT PROCESSES

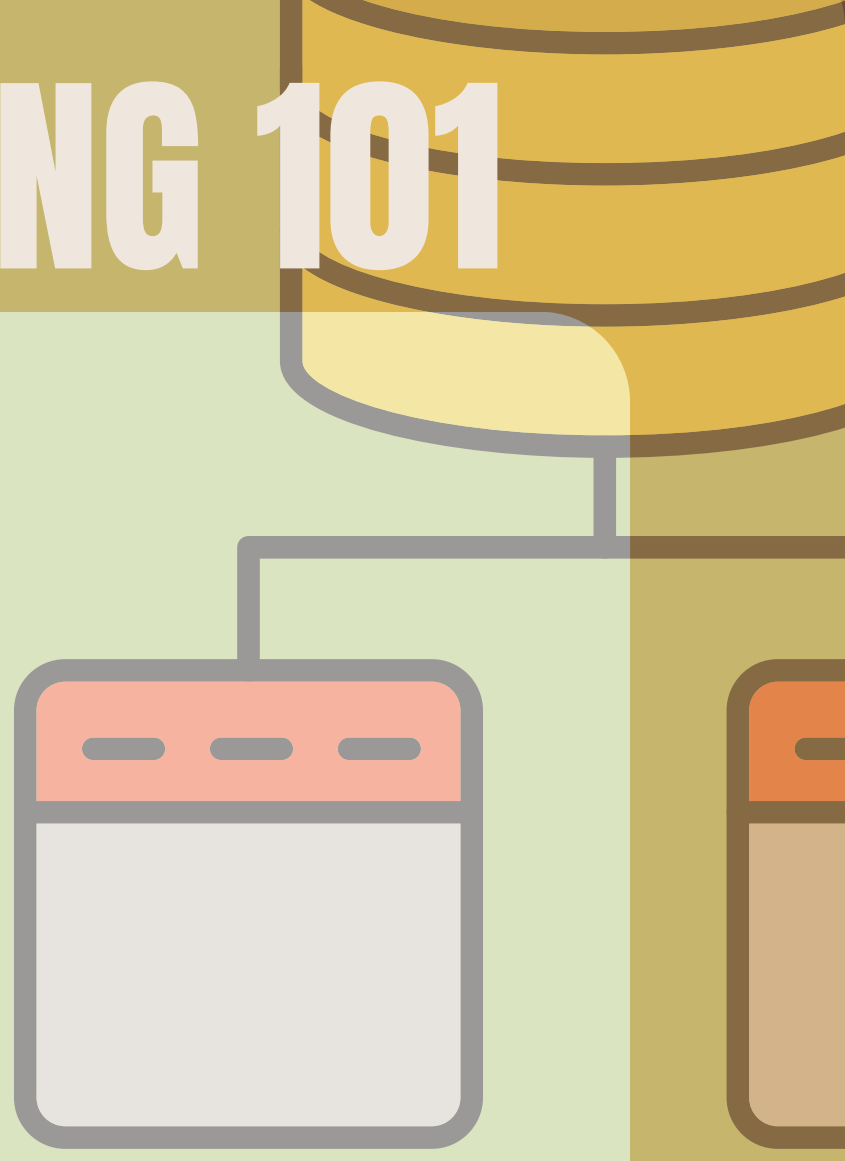
Extract, Transform, Load (ETL) or Extract, Load, Transform (ELT) for data integration.

Using Azure Data Factory to ETL data from on-prem SQL Server to Azure Synapse Analytics.



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DATA MODELING

Designing data models such as star schema and snowflake schema to optimize for queries.

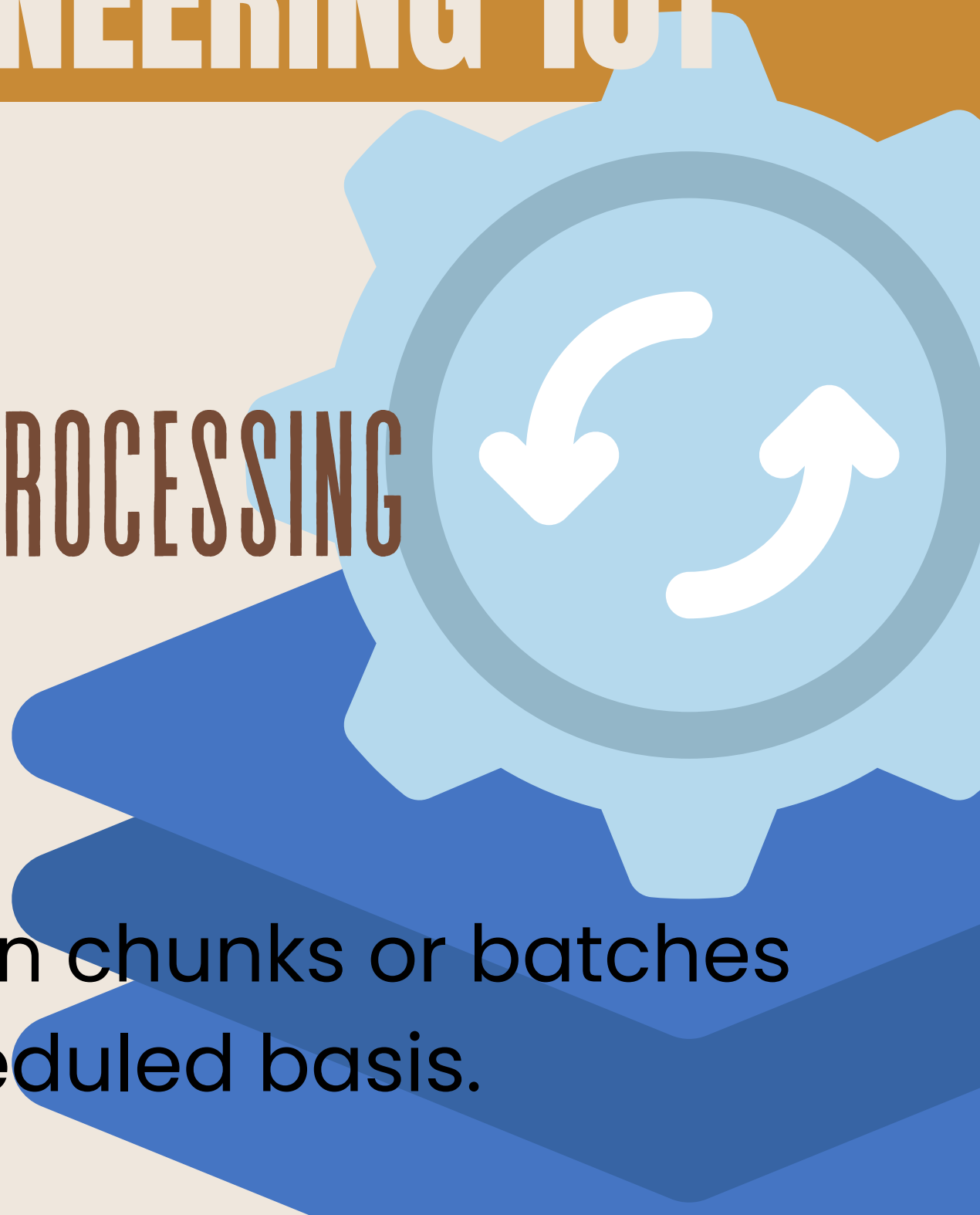
Creating star schemas for a retail company to model sales and products.



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BATCH PROCESSING



Processing data in chunks or batches on a scheduled basis.

Using Apache Spark to process sales data from yesterday's transactions.



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STREAM PROCESSING

Processing data in real time or near-real time as it's produced.

Using Apache Kafka with Apache Flink for real-time fraud detection on transaction data.



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DATA LAKES

Centralized repositories for storing structured, semi-structured, and unstructured data at scale.

Building a data lake using Azure Data Lake to store IoT data, social media feeds, and logs.



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DATA LAKEHOUSE

Architecture that combines data lakes and data warehouses for unified analytics.

Using Delta Lake to enable both batch and real-time analytics on the same data in Azure.



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DATA PARTITIONING

Splitting large datasets into smaller partitions to improve query performance and manageability.

Partitioning S3 bucket files by year, month, day for better query performance using AWS Athena.



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DATA SHARDING

Breaking down large datasets horizontally across multiple databases to improve scalability.

Sharding user data across multiple PostgreSQL instances to distribute the load.



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DATA PIPELINES

Automated workflows that move data between systems in a scheduled or triggered manner.

Building an Airflow DAG to automate the ETL process for daily sales data.



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DATA QUALITY

Ensuring accuracy, completeness, consistency, and validity of data.

Implementing data validation rules to check null values or duplicates using Great Expectations.



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DATA LINEAGE

Tracking the flow of data from source to destination for audit and troubleshooting.

Using tools like Apache Atlas or DataHub to track data lineage in an ETL pipeline.



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SCHEMA EVOLUTION

The ability to adapt to changes in the structure of data sources and their schemas.

Handling new columns in a table in a Spark job without breaking the existing pipeline.



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CHANGE DATA CAPTURE (CDC)

Capturing and tracking changes in source data for real-time updates.

Using Debezium to track changes in MySQL database and load them into a Kafka topic.



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DATA GOVERNANCE

Defining rules, policies, and standards for managing and accessing data.

Implementing data governance with data catalog tools like Collibra or Alation.



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DATA SECURITY

Protecting data at rest, in transit, and ensuring secure access.

Encrypting sensitive data using AES-256 while stored in an AWS S3 bucket.



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DATA ANONYMIZATION

Removing personally identifiable information to maintain user privacy.

Masking user IDs and phone numbers before sending them to the analytics team.



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METADATA MANAGEMENT

Storing and managing information about the data, like source, structure, and usage.

Using Hive Metastore to manage metadata for tables in an Apache Hadoop cluster.



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SCALABLE STORAGE

Designing storage solutions to handle growing data volumes.

Using Amazon S3 or Azure Blob Storage for scalable storage of raw data files.



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COLUMNAR STORAGE

Storing data in columnar format for analytics use, improving performance.

Using Parquet file format for efficient data processing with Apache Spark.



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DATA NORMALIZATION

Organizing data to reduce redundancy and improve integrity.

Structuring customer data in a 3NF schema in an RDBMS to eliminate redundancy.



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NOSQL DATABASES

Non-relational databases for storing semi-structured or unstructured data.

Using MongoDB to store JSON documents for product catalog data.



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DISTRIBUTED SYSTEMS

Systems that divide tasks among multiple machines for performance and availability.

Using Hadoop Distributed File System (HDFS) for storing petabytes of data across nodes.



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DATA REPLICATION

Creating redundant copies of data for failover and availability.

Using Azure Cosmos DB's geo-replication for ensuring high availability across regions.



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DATA BACKUP

Creating backup copies of data for disaster recovery purposes.

Scheduling PostgreSQL backups using AWS RDS automated backup feature.



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LOAD BALANCING

Distributing data processing workloads across nodes to avoid bottlenecks.

Using Kubernetes to distribute data processing pods in a Spark cluster.



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WORKFLOW ORCHESTRATION

Managing dependencies, scheduling, and monitoring data pipeline workflows.

Using Apache Airflow to orchestrate data pipeline tasks with DAGs.



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DATA VALIDATION

Testing data for expected values and types before processing.

Writing data validation scripts in Python to verify incoming data for missing or incorrect values.



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DATA TRANSFORMATION

Converting raw data into a usable format (e.g., aggregation, joins).

Using PySpark to join user and transaction data for downstream analysis.



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OLAP VS. OLTP

OLAP is used for analytics, OLTP for day-to-day transaction processing.

Using OLAP for querying historical sales data and OLTP for processing real-time purchase orders.



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SQL QUERY OPTIMIZATION

Improving SQL queries for better performance and faster execution.

Adding indexes and rewriting SQL joins to optimize query performance in a relational database.



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SERVERLESS DATA PROCESSING

Processing data without managing underlying servers.

Using AWS Glue for running serverless ETL jobs on demand.



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EVENT-DRIVEN PROCESSING

Reacting to events to trigger data processing workflows.

Using AWS Lambda to trigger data transformation based on new S3 object uploads.



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JOB SCHEDULING

Automating the execution of jobs at specific intervals.

Scheduling nightly ETL jobs using Azure Data Factory's pipeline scheduling feature.



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DATA SCHEMA REGISTRY

Storing schema definitions for data to facilitate consistent data exchange.

Using Confluent Schema Registry for managing Avro schemas in Kafka topics.



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DATA VERSIONING

Keeping track of changes to data and datasets for reproducibility.

Versioning datasets using Delta Lake to track changes to the data over time.



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DATA CATALOGING

Indexing data assets to make them searchable and discoverable.

Using Azure Purview to catalog all datasets in a data warehouse environment.



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DATA INGESTION

Pulling in data from different data sources for storage or processing.

Using Apache NiFi for ingesting sensor data into an HDFS data lake.



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DATA LAKE STORAGE

Using data lakes for flexible data storage, especially for unstructured data.

Storing IoT sensor data in Azure Data Lake for processing later with Spark.



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DATA CLEANSING

Correcting or removing corrupt, incomplete, or inaccurate data.

Removing null values and outliers from customer data using Python pandas.



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MASTER DATA MANAGEMENT

Creating a single, trusted source of key business data.

Maintaining a unified view of customer data using Informatica MDM.



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CLOUD DATA SERVICES

Leveraging cloud services for data storage and processing.

Using AWS Redshift for data warehousing and Azure Synapse for data analytics.



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